

LOK JAGRUTI UNIVERSITY (LJU)

INSTITUTE OF ENGINEERING AND TECHNOLOGY

Department of CE/IT/CSD/AIML/AIDS /CSE/CST/CS&IT/CEA/MA&CP/CSE(AI)/ CSE(DS)/CSE(CS)

(701,702,703,704,705, 712,713,714,715,716,717,718,719)

Bachelor of Engineering (B.E.) – Semester – I

Course Code:	117011191/117021191/117031191/117041191/117051191/117121191/117131191/117141191/117151191/017161191/017171191/017181191/017191191
Course Name:	Mathematics - I
Category of Course:	Basic Science Course (BSC)
Prerequisite Course:	---

Teaching Scheme				
Lecture (L)	Tutorial (T)	Practical (P)	Credit	Total Hours
4	2	0	6	60

Syllabus				
Unit No.	Topic	Prerequisite Topic	Successive Topic	Teaching Hours
01	Basic Algebra			2 (2%)
	1.1 Indices	---	---	
	1.2 Surds	---	---	
	1.3 Expansions and factorization	---	---	
	1.4 Logarithm and Exponential Function with its Application	---	---	
02	Trigonometry and Geometry			2 (3%)
	2.1 Angles	---	---	
	2.2 Trigonometric functions of sum and difference of two angles	---	---	
	2.3 Law of sines and cosines	---	---	
03	Basic Differentiation			3 (4%)
	3.1 Basic differentiation by formulae	---	---	
	3.2 Product and quotient rule	---	---	
	3.3 Chain rule and composite function	---	---	
	3.4 Roll's theorem	Basic differentiation (017011191-Unit-3)	---	
	3.5 Lagrange's theorem	Basic differentiation (017011191-Unit-3)	---	
	3.6 Cauchy's mean value theorem	First and higher order partial derivatives (017011191-Unit-3)	---	
	3.7 Indeterminate forms and L'Hospital's rule	---	---	
04	Partial Derivatives			6 (9%)
	4.1 Functions of several variables	---	---	
	4.2 Geometric interpretation of partial derivatives	---	---	
	4.3 Limits and continuity of function of several variables	Basic differentiation (017011191-Unit-3)	---	
	4.4 First and higher order partial derivatives	---	---	
	4.5 Euler's theorem and modified Euler's theorem	Basic differentiation (017011191-Unit-3)	---	
	4.6 Total derivatives and chain rule	First and higher order partial derivatives (017011191-Unit-3)	---	
	4.7 Implicit function	---	---	
05	Application of Partial Derivatives			4 (7%)
	5.1 Tangent plane and normal line	First order partial derivatives(017011191-Unit-3)	---	
	5.2 Total differentiation and approximation		---	
	5.3 Extreme values	First and higher order partial derivatives(017011191-Unit-3)	---	
	5.4 Method of Lagrange multipliers.	---	---	
	5.5 Jacobian	First order partial derivatives(017011191-Unit-3)	---	
06	Multiple Integral			15 (25%)
	6.1 Basic integration by formulae	---	---	
	6.2 Integration by parts	---	---	
	6.3 Double integral over rectangles and general regions	Basic integration(017011191-Unit-3) Double integral over rectangles and general regions (017011191-Unit-7)	---	
	6.4 Change of order of integration	Basic integration(017011191-Unit-3) Double integral over rectangles and general regions (017011191-Unit-7)	---	
	6.5 Double integration in polar coordinates	Basic integration(017011191-Unit-3)	---	

	6.6 Change of variables in double integration by Jacobian	---	---	
	6.7 Triple integration	---	---	
	6.8 Area enclosed by plane curve using double integration	Double integral over rectangles and general regions, Double integration in polar coordinates (017011191-Unit-7)	---	
	6.9 Triple integration in cylindrical and spherical co-ordinates	Triple integration (017011191-Unit-7)	---	
07	Vector Calculus-I			7 (13%)
	7.1 Vector and its properties	---	---	
	7.2 Parametrization of curves	---	---	
	7.3 Arc length of curve in space	Basic integration (017011191-Unit-3) First order partial derivatives(017011191-Unit-3)	---	
	7.4 Gradient of a scalar point function and surface normal vector	---	Models: Linear Modeling (017012391-Unit -4), Stochastic Gradient Descent (017012391-Unit -4)	
	7.5 Directional derivatives	Gradient(017011191-Unit-9)	---	
	7.6 Divergence of vector field	First order partial derivatives(017011191-Unit-3)	---	
	7.7 Curl of vector field	---	---	
	7.8 Scalar potential function of conservative field	---	---	
08	Vector Integral			7 (12%)
	8.1 Line integral	Basic integration (017011191-Unit-3)	---	
	8.2 Work done	Line integral (017011191-Unit-10)	---	
	8.3 Circulation and Flux	Line integral (017011191-Unit-10)	---	
	8.4 Green's theorem in the plane (without proof)	Line integral (017011191-Unit-10), Double integral over rectangles and general regions (017011191-Unit-8)	---	
09	Graph Theory			8 (15%)
	9.1 Introduction to Graphs and Definitions	---	---	
	9.2 Path and Circuits	---	---	
	9.3 Cut Sets and Cut Vertices	---	---	
	9.4 Graph Representations and Matrix Theory			
	9.5 Graph Coloring, Chromatic Polynomial and Matching	---	---	
10	Trees			6 (10%)
	10.1 Mathematical Foundations of Trees	Carrier Generation (creation of EHP) and carrier recombination (017011192 -Unit-6.3)	---	
	10.2 Spanning Trees	---	---	

Course Outcome	
	<i>Upon completion of the course students will be able to</i>
CO1	Solve problems of differentiation, integration, trigonometry and some practical problems, such as constrained optimization problems and other problems involving Partial differentiation and to calculate directional derivatives and gradients.
CO2	Evaluate a double integral in polar coordinates and triple integral to find area and volume in rectangular coordinates, cylindrical coordinates, and spherical coordinates.
CO3	Apply gradient to solve problems involving normal vectors to level surfaces and to Explain the concept of a vector integration in a plane(2-dimensions) and in the space(3-dimensions).
CO4	Understand the concepts of Graph theory in context of computer science and to solve problems related to nonlinear structures like Tree
Suggested Reference Books	
1	Calculus with Early Transcendental Functions, James Stewart, Cengage Learning.
2	Thomas’ Calculus, Maurice D. Weir, Joel Hass, Frank R. Giordano, Pearson Education.
3	Higher Engineering Mathematics, B.S.Grewal, Khanna Publishers.
4	Advanced Engineering Mathematics, Erwin Kreysig, Wiley Publication.
5	Graph Theory with Applications to Engineering & Computer Science, Narsingh Deo, Dover Publications, INC.Mineola, New York

List of Open Source Software/Learning website	
1	http://nptel.ac.in/

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Bachelor of Engineering (B.E.) - Semester - I

Course Code:	017012194/017022194/017032194/017042194/017052194/017122194/017132194/017142194/017152194/017162194/017172194/017182194/017192194
Course Name:	Computer Programming using Python-I
Category of Course:	Engineering Science Course (ESC)
Prerequisite Course:	Mathematics - I (117011191)

Teaching Scheme				
Lecture (L)	Tutorial (T)	Practical (P)	Credit	Total Hours
3	0	6	6	30

Syllabus				
Unit No.	Topic	Prerequisite Topic	Successive Topic	Teaching Hours
01	Introduction to Computer			1 (2%)
	1.1 Basic Model of Von-Neumann Architecture	---	---	
	1.2 Programming Language - Machine, Assembly and High-Level Languages, Assembler, Compiler, Interpreter			
	1.3 Algorithm and Flowchart			
02	Introduction to Python and Jupyter Notebooks			3 (10%)
	2.1 Basic Data Types -integers, Booleans, floats, strings, type() function, typecasting	Programming Language (017012194 – Unit-1)	Introduction to Data Structure (117013292 - Unit 1)	
	2.2 Declaring and using variables, Comments, Reading input from users, printing output	Basic Data Types (017012194 – Unit-2)	---	
	2.3 Arithmetic Operators: +, -, *, /, **, //, % Logical Operators: and, or, not Membership Operators: in, not in Assignment Operators: =, +=, -=, *=, /=, **=, %= Comparison Operators: ==, !=, >, <, >=, <= Bitwise Operators - &, , ^, ~, <<, >> Ternary Operators, Operator precedence, ord and chr function, math library	Basic Data Types (017012194 – Unit-2)	---	
	2.4 Jupyter Notebooks: Creating, Opening, Saving and Downloading Notebooks, running cells		---	
03	Conditional Execution and Iterations			4 (12%)
	3.1 Control Statements: Simple if, if-else, if-elif-else, Nested if	Algorithm and Flowchart (017012194 – Unit-1), Operators (017012194 – Unit-2) Loops (017012194 – Unit-3)	---	
	3.2 Loops: For using range function with else, while loop with else, Nested loop			
	3.3 Break, Continue, Pass Statement			
04	Functions, Scoping and Abstraction			3 (12%)
	4.1 Declaring, Defining and Invoking a Function, Function Specification	Conditional Execution and Iterations (017012194-Unit-3)	Introduction to Object-Oriented Programming (Python-II Unit-1)	
	4.2 Function arguments: keyword, default, positional, variable-length		Stack (117013292 - Unit 2)	
	4.3 Local v/s Global variables, Nested Functions, Recursion			
05	Immutable Data Structures			4 (14%)
	5.1 Strings -immutability, declaring, accessing through for loop, slicing, concatenation, len, capitalize(), enumerate(), isalnum(), isalpha(), islower(), isupper(), lower(), upper(), isnumeric(), find(), index(), split(), strip(), removing punctuation using translate(), count()	Functions (017012194-Unit-4)	---	
	5.2 Tuples -immutability, create, assign, access, del, slicing, concatenation, comparing tuples using >, <, ==, count() function, Generators (yield)			
	5.3 Built-in functions -sorted, reversed, min, max			
06	Mutable Data Structures			5 (18%)
	6.1 Lists - mutability, declaring, indexing and slicing, accessing through for loop, nested lists, append(), count(), extend(), index(), insert(), pop(), remove(), reverse(), sort(), len(), max(), min(), sum(), list comprehension	Tuples (immutability, slicing, access) (017012194-Unit-5)	---	
	6.2 Dictionary - create, accessing elements, accessing using for loop, nested dictionaries, membership operations, clear(), copy(), get(), items(), keys(), update(), values(), dictionary comprehension	Lists (iteration, comprehension syntax, mutability concepts) (017012194-Unit-6)		
	6.3 Sets - declaring, properties of sets, accessing using for loop, membership operations, set(), frozenset(), issubset(), issuperset(), union(), intersection(), difference(), symmetric_difference(), copy(), set comprehension	---		

	6.4 Lambda functions - anonymous functions, syntax of lambda, use of lambda with map(), filter(), reduce() functions	Functions (017012194-Unit-4)		
07	Modules, Directories			1 (5%)
	7.1 Modules and Packages -difference between import and from...import	Lambda functions (017012194-Unit 06)	---	
	7.2 Python OS Module -getcwd(), chdir(), mkdir(), listdir(), remove(), rmdir(). Time Module - time(), sleep(), ctime() Datetime Module - today(), now(), strftime(), strptime(), replace()	Modules and Packages (017012194-Unit 07)		
	7.3 Random Module - randint(), random(), choice(), shuffle() Math Module - sqrt(), pow(), ceil(), floor(), factorial(), constants			
08	Regular Expressions			2 (5%)
	8.1 String formatting using f-strings	---	f-strings (017012194-Unit 08), Modules and Packages (017012194 -Unit 07), Strings (017012194-Unit 05)	
	8.2 Introduction to Regular Expressions - pattern matching, re module, match(), search(), findall(), sub()			
	8.3 Regular Expression Patterns - metacharacters, character classes, predefined sets, quantifiers, anchors, grouping			
	8.4 Working with Regular Expressions - extraction, validation, replacement, splitting using regex			
09	Working with Files			3 (10%)
	9.1 Working with text files, opening files -open in w, r and a modes with open function	Working with Regular Expressions (017012194-Unit 08)	---	
	9.2 Reading data from files -read(), readline(), readlines(), seek() and tell() functions, writing data to files -write(), writelines(), closing files			
	9.3 Working with JSON files - introduction to JSON, difference between text and JSON files, importing json module, reading JSON data using load() and loads()			
	9.4 Writing JSON data using dump() and dumps(), conversion between Python objects and JSON, handling basic JSON structures (list and dictionary)			
10	Arrays and Plotting using NumPy & Matplotlib			4 (12%)
	10.1 Arrays - creating NumPy ndarray object using array(), 1D, 2D and 3D arrays, data types (dtype), array attributes (ndim, size, shape)	Writing JSON data (017012194-Unit 09)	---	
	10.2 Array Indexing, Slicing, Shape, Reshaping, iteration, negative indexing, flattening arrays (flatten(), ravel())	Arrays (017012194-Unit 10)	---	
	10.3 Built-in functions - zeros(), ones(), arange(), concatenate(), array_split(), where(), sort(), random	Indexing/Slicing/Reshaping (017012194-Unit 10)	---	
	10.4 Array Operations - element-wise operations, arithmetic operations on arrays, aggregation functions (sum(), mean(), min(), max())	Built-in functions (017012194-Unit 10)	---	
	10.5 Basic plotting using Matplotlib: line, bar, scatter, histogram, pie, area	Array Operations (017012194-Unit 10), Modules and Packages (017012194-Unit 7)	Data Visualization (Python-II Unit-6)	
	10.6 Customization of plots: labels, title, legend, colors, grid, Subplots and multiple plots	Basic plotting (017012194-Unit 10)		

Sr No.	Practical Title	Link to Theory Syllabus
1	Write an algorithm and draw a flowchart to find the largest among three numbers. Also write a separate algorithm and flowchart to compute the factorial of a given number.	Unit-1
2	Write a Python program that reads the radius of a circle and a temperature in Celsius from the user, then computes and prints (i) the area and circumference of the circle using the math library and (ii) the temperature converted to Fahrenheit. Demonstrate the use of arithmetic operators, typecasting and the type() function on the inputs.	Unit-2
3	Take three variables and get their values from the user. Without using any sorting functions or Python built-in data structures (lists, tuples, dictionaries, etc.), sort these numbers in ascending and descending orders.	Unit-3
4	Calculate the number of basic American coins given a value less than 1 dollar. A penny is worth 1 cent, a nickel is worth 5 cents, a dime is worth 10 cents, and a quarter is worth 25 cents. It takes 100 cents to make 1 dollar. So given an amount less than 1 dollar (if using floats, convert to integers for this exercise), calculate the number of each type of coin necessary to achieve the amount, maximizing the number of larger denomination coins. For example, given \$0.76, or 76 cents, the correct output would be “3 quarters and 1 penny.” Output such as “76 pennies” and “2 quarters, 2 dimes, 1 nickel, and 1 penny” are not acceptable.	Unit-3
5	Write a Python program that prompts the user to enter numbers and stops only when the user enters “done”. After this, print the average of the numbers and the minimum number and the maximum number from among all the numbers entered by the user. Note: You are not allowed to use any built-in structures like lists, tuples, etc. or any built-in functions like max, min, etc. E.g., if the user enters: 3, 4, 1, 6 ,“done” The program should print the Average: 3.5 Minimum Number: 1 Maximum Number: 6	Unit-3
6	Given an integer value, return a string with the equivalent English text of each digit. For example, an input of 89 results in “eighty-nine” being returned. Restrict values to be between 0 and 1,000.	Unit-3
7	Write a Python program to check whether the given no is Armstrong or not using user defined function.	Unit-4
8	Write a python program to implement isPalindrome() function to check given string is palindrome or not.	Unit-4
9	Write a Python program to find reverse of given number using user defined function.	Unit-4
10	a) Create a function called findchr(), with the following declaration: def findchr(string, char) findchr() will look for character char in string and return the index of the first occurrence of char, or -1 if that char is not part of string. You cannot use string.find() or string.index() functions or methods. (b) Create another function called rfindchr() that will find the last occurrence of a character in a string. Naturally this works similarly to findchr(), but it starts its search from the end of the input string. (c) Create a third function called subchr() with the following declaration: def subchr(string, origchar, newchar) subchr() is similar to findchr() except that whenever origchar is found, it is replaced by newchar. The modified string is the return value. Use of any substring or replace built-in string methods is prohibited.	Unit-4
11	Create a function that will return another string similar to the input string, but with its case inverted. For example, input of “Mr. Ed” will result in “mR. eD” as the output string. Note: Use of the built-in swapcase function is prohibited.	Unit-4
12	Write a Python program that accepts a sentence from the user and, using string methods, prints the sentence in upper and lower case, the count of alphabetic and numeric characters, and the sentence with all punctuation removed. Split the sentence into words and, using a tuple, display each word together with its length. Also use sorted(), min() and max() to display the words in alphabetical order and to find the shortest and longest word.	Unit-5
13	Write a Python program which will return the sum of the numbers in the list, returning 0 for an empty list. Except the number 13 is very unlucky, so it does not count and number that come immediately after 13 also do not count. Example : [1, 2, 3, 4] = 10 [1, 2, 3, 4, 13] = 10 [13, 1, 2, 3, 13] = 5	Unit-6
14	Write a Python program which takes a list and returns a list with the elements "shifted left by one position" so [1, 2, 3] yields [2, 3, 1]. Example: [1, 2, 3] → [2, 3, 1] [11, 12, 13] → [12, 13, 11]	Unit-6
15	Write a program to ask the user to input a list of names, in the format “Last Name, First Name,” i.e., last name, comma, first name. Write a function that manages the input so that when/if the user types the names in the wrong order, i.e., “First Name Last Name,” the error is corrected, and the user is notified. This function should also keep track of the number of input mistakes. When the user is done, sort the list, and display the sorted names in “Last Name, First Name” order.	Unit-6
16	Given a list L of size N. You need to count the number of special elements in the given list. An element is special if removal of that element makes the list balanced. The list will be balanced if sum of even index elements is equal to the sum of odd index elements. Example Input Input 1:	Unit-6

	A = [2, 1, 6, 4] Input 2: A = [5, 5, 2, 5, 8] Example Output Output 1: 1 Output 2: 2 Explanation 1: After deleting 1 from list : [2,6,4] (2+4) = (6) Hence 1 is the only special element, so count is 1 Explanation 2: If we delete A[0] or A[1] , list will be balanced (5+5) = (2+8) So A[0] and A[1] are special elements, so count is 2.	
17	Explain the use of the following functions with code and output: getcwd(), chdir(), mkdir(), listdir(), remove(), rmdir()	Unit-7
18	Write a Python program that uses the re module to validate and extract information from user-entered text: (i) check whether an entered string is a valid email address, (ii) extract all mobile numbers and dates from a given paragraph using findall(), and (iii) replace every sequence of one or more whitespace characters with a single space using sub(). Display all the results using f-strings.	Unit-8
19	Create a Python program to read a text file and do following: 1. Print no. of lines 2. Print no. of unique words 3. Store each word with its occurrence in dictionary.	Unit-9
20	Write a python program that reads a text file containing full-name of the employees of a company and then from these names it extracts Last-name of employee and stores them in sorted order in new file. Input Text file contains names in format: First-name Middle-name Last-name	Unit-9
21	Write a python program to read line by line from a given files file1 & file2 and write into file3.	Unit-9
22	Write a “pager” program. Your solution should prompt for a filename, and display the text file 25 lines at a time, pausing each time to ask the user to enter the word “continue”, in order to show the next 25 lines or enter the word “stop” to close the file.	Unit-9
23	Using NumPy, create a 3x3 two-dimensional ndarray from user-entered values and display its ndim, size, shape and dtype. Then demonstrate array slicing and reshaping, generate arrays using zeros(), ones() and arange(), perform element-wise arithmetic on two arrays, and compute the sum, mean, minimum and maximum of the array using NumPy aggregation functions.	Unit-10
24	Write a Python program to plot the monthly sales data of a shop using a line chart , and the number of units sold by 5 different products using a bar chart , using Matplotlib.	Unit-10
25	Write a Python program that takes the marks of 30 students as input and plots (a) a scatter plot of Marks vs. Roll Number, and (b) a histogram showing the distribution of marks, using Matplotlib.	Unit-10
26	Write a Python program to create a figure with multiple subplots (2x2 layout) displaying line, bar, scatter and pie charts of related data side-by-side, with appropriate titles and legends for each subplot.	Unit-10

Major Components/ Equipment	
Sr. No.	Component/Equipment
1	Computer
2	Jupyter Notebook

Course Outcome	
	<i>Upon completion of the course students will be able to</i>
CO1	Explain the basic model of computer architecture and the role of language translators, and write, execute, and debug simple Python programs in Jupyter Notebooks using appropriate data types, operators, conditional statements, and loops.
CO2	Design and implement modular programs using functions with different types of arguments, scoping rules, and recursion, and process immutable data structures such as strings and tuples effectively.
CO3	Manage and manipulate mutable data structures such as lists, dictionaries, and sets along with lambda functions, and organize programs using Python modules, packages, and directory-handling operations.
CO4	Apply regular expressions for pattern matching and text processing, perform file handling with text and JSON files, and carry out numerical computations using NumPy arrays.
Suggested Reference Books	
1	John V Guttag; Introduction to Computation and Programming Using Python; Prentice Hall
2	Wesley Chun; Core Python Programming; Prentice Hall
3	Kenneth A. Lambert; Fundamentals of Python - First Programs; CENGAGE Publication
4	Robert Johansson; Numerical Python: Scientific Computing and Data Science Applications with Numpy, SciPy and Matplotlib; Apress
5	R. Nageswara Rao; Core Python Programming; dreamtech press

List of Open Source Software/Learning website	
1	https://docs.python.org/3/tutorial/
2	https://numpy.org/doc/stable/
3	https://www.w3schools.com/python/
4	https://www.coursera.org/specializations/python

Practical Project/Hands on Project		
Sr. No.	Project List	Linked with Unit
1	Menu-Driven Contact Book (File & JSON based): Develop a console-based contact book that stores each contact as a dictionary with name, phone number and email address, and keeps all contacts in a JSON file so that data is retained between runs. Provide user-defined functions to add a new contact, view all contacts, search a contact, modify a contact and delete a contact. Load and save the data using the json module (load()/loads() and dump()/dumps()). Validate the phone number and email address using the re module before storing them.	Unit-4, 6, 8, 9
2	Student Marks Analyzer using NumPy: Read the marks of a class of students for several subjects from a text file into a NumPy two-dimensional array. Using NumPy aggregation functions (sum(), mean(), min(), max()), compute each student’s total and average marks and each subject’s highest, lowest and mean marks. Assign a grade to every student using conditional statements and write a neatly formatted result summary into a new text file.	Unit-3, 9, 10
3	Text File Word-Frequency & Report Generator: Take a text file as input, read its contents, convert the text to lower case and remove punctuation using string methods, then split it into words and build a dictionary of word frequencies. Display the total number of words, the number of unique words and the top-N most frequent words (sorted using a lambda key). Use a regular expression to count how many words match a pattern entered by the user, and finally write the complete report to an output file.	Unit-5, 6, 8, 9
4	Command-Line Calculator & Unit Converter: Build a menu-driven program that repeatedly accepts a choice from the user until they choose to exit. It should perform arithmetic operations (using operators and the math module functions such as sqrt(), pow(), factorial(), ceil() and floor()) and length, weight and temperature conversions, each implemented as a separate user-defined function using default and keyword arguments. Log every calculation together with a timestamp (using the datetime module) into a history text file.	Unit-2, 3, 4, 7, 9
5	Simple Quiz Game with Scorecard: Load a set of multiple-choice questions from a JSON file into a list of dictionaries. Present the questions to the user in a random order using the random module (shuffle()), check each answer, and keep the score. At the end, display the score, the percentage obtained and a pass/fail message. Append each player’s name, score and date to a leaderboard text file and display the leaderboard sorted with the highest scores first.	Unit-3, 6, 7, 9
6	Sales & Performance Dashboard using NumPy and Matplotlib: Read a shop's daily sales data (product-wise units sold across a week) from a text file into a NumPy 2D array. Using NumPy functions, compute total units sold per product, per day, and overall weekly revenue (sum(), mean(), max(), min()). Use where() to identify the best and worst performing products, and sort() to rank products by total sales. Visualize the results using Matplotlib: a line chart for daily revenue trend, a bar chart for product-wise total sales, and a pie chart for revenue share by product — each customized with titles, labels, legends, colors and grid, arranged as subplots in a single dashboard figure.	Unit-3, 9,10

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Bachelor of Engineering (B.E.) – Semester – I

Course Code:	017013191/017023191/017033191/017043191/017053191/017123191/ 017133191/017143191/017153191/017163191/017173191/017183191/ 017193191	Teaching Scheme				
Course Name:	Software Engineering	Lecture (L)	Tutorial (T)	Practical (P)	Credit	Total Hours
Category of Course:	Professional Core Course (PCC)	3	0	2	4	20
Prerequisite Course:	-					

Syllabus				
Unit No.	Topic	Prerequisite Topic	Successive Topic	Teaching Hours
01	Introduction to Software Engineering and Software Process Models			3 (15%)
	1.1 Basic Concepts of Software Engineering, Software Development Life-Cycle.	---	---	
	1.2 Study of Different Software Process Models, The Linear Sequential Model, The Prototyping Model, The RAD Model, Spiral Model.			
02	Agile Development			1 (5%)
	2.1 Agile Process, Extreme Programming.	---	---	
	2.2 Scrum Model	---		
03	Requirement Analysis and Specification			2 (10%)
	3.1 Requirement Gathering and Analysis, Feasibility Analysis.	---	---	
	3.2 Software Requirement Specification (SRS).	---		
04	Function Oriented Analysis and Design			1 (5%)
	4.1 Data Modelling Concepts	---	---	
	4.2 Data Flow Model (DFD Diagram).			
05	Object Oriented Analysis and Design			3 (15%)
	5.1 Use-Case Diagram, Activity diagram, Class diagram	----	---	
	5.2 State diagram, sequence diagram			
06	Software Project Management			3 (15%)
	6.1 Process and Product Metrics (Size Oriented and Function Oriented).	---	---	
	6.2 Empirical Estimation Model-COCOMO Model.	---		
	6.3 Project Scheduling and Tracking	---		
07	Testing Types			2 (10%)
	7.1 Software Testing Fundamentals	---	---	
	7.2 Unit Testing, Integration, Acceptance Testing, Validation Testing and System Testing.			
08	Testing Techniques			2 (10%)
	8.1 Test Case Design.	---	---	
	8.2 White-Box Testing and Black-Box Testing.	---		
09	Software Quality Assurance			2 (10%)
	9.1 Software Quality Assurance.	---	---	
	9.2 The ISO9000 Quality Standards, Software Reliability.			
	9.3 CMM			
10	Software Configuration and Maintenance Management			1 (5%)
	10.1 SCM Process.	---	---	
	10.2 Version Control, Change Management.	---		

Major Components/ Equipment	
Sr. No.	Component/Equipment
1	Computer
2	Microsoft Visio, Testing tool, Project Management tool (Cloud based or desktop based),Version Control Tool(GIT,SVN or any other)

Sr No.	Practical Title	Link to Theory Syllabus
Choose any one project and do the following exercises with respect to selected project definition. A. Hotel Management System B. Library Management system C. Hostel Management System D. Blood bank Management System E. Laboratory Management System F. College Management System G. Online Furniture Selling platform for single vendor H. Online Clothes Selling platform for multiple vendor I. Vendor Management system for E-commerce System J. Online Job Portal K. Online Books Auction L. Art gallery Management System. M. HR Management System N. Stock Management Portal O. Inventory Management System		
1.	Write the Feasibility study and accordingly decide the complete problem statement of your chosen Project .	Unit – 3
2.	What are the factors that influence the choice of SDLC models? Analyze each of them for your chosen Project and decide which SDLC model is most suitable.	Unit – 1
3.	Which are other models apart from SDLC models? Is any of them suitable for your chosen Project ? If not, Justify.	Unit – 2
4.	Identify the requirement development activities for your chosen Project . Also specify list of functional and non-functional requirements for the same.	Unit – 3
5.	Identify different modules for your chosen Project along with their detailed description.	Unit – 3
6.	Draw the DFD-Level 0 for your chosen Project	Unit – 4
7.	Draw the DFD-Level 1 for your chosen Project	Unit – 4
8.	Draw the DFD-Level 2 for your chosen Project	Unit – 4
9.	Describe the user scenarios for your chosen Project with pre and post conditions.	Unit – 5
10.	Draw the use case diagram for your chosen Project	Unit – 5
11.	Draw the state diagram for your chosen Project	Unit – 5
12.	Draw the class diagram for your chosen Project	Unit – 5
13.	Draw the sequence diagram for your chosen Project	Unit – 5
14.	Draw the activity diagram for your chosen Project	Unit – 5
15.	Online loan system has two modules for the two basic services, namely Car loan service and House loan service. The two modules have been named as Car_Loan_Module and House_Loan_Module. Car_Loan_Module has 2000 lines of uncommented source code. House_Loan_Module has 3000 lines of uncommented source code. Car_Loan_Module was completely implemented by Mike. House_Loan_Module was completely implemented by John. Mike took 100 person hours to implement Car_Loan_Module. John took 200 person hours to implement House_Loan_Module. Mike’s module had 5 defects. John’s module had 6 defects. With respect to the context given, which among the following is an INCORRECT statement? Choose one: 1. John’s Quality is better than Mike’s Quality 2. John’s Productivity is more than Mike’s Productivity 3. John introduced more defects than Mike 4. John’s Effort is more than Mike’s Effort Also Calculate size-oriented metrics for your chosen Project.	Unit – 6
16.	Compute the function point, productivity, documentation, cost per function for the following data: 1. Number of user inputs = 24 2. Number of user outputs = 46 3. Number of inquiries = 8 4. Number of files = 4 5. Number of external interfaces = 2 6. Effort = 36.9 p-m 7. Technical documents = 265 pages 8. User documents = 122 pages 9. Cost = \$7744/ month Various processing complexity factors are: 4, 1, 0, 3, 3, 5, 4, 4, 3, 3, 2, 2, 4, 5. Also Calculate Function Oriented metrics for your chosen Project.	Unit – 6
17.	Suppose a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three model i.e., organic, semi-detached & embedded. Also using COCOMO model calculate effort and development time for your chosen Project.	Unit – 6

18.	Show how project scheduling is carried out for your chosen Project using open source framework – GanttProject (Or any other tool of your choice)	Unit – 6																																	
19.	Show how project management is carried out for your chosen Project using – JIRA (Or any other tool of your choice)	Unit – 6																																	
20.	In the context of the above defect categories, classify the following statements under the defect categories and mention in the table given below. 1. Divide by Zero Error is not guarded 2. Usage of 3.14 in the statement Circle_Area = 3.14 * Radius * Radius; 3. 3500 lines of code in a single function 4. A pointer is declared but not initialized. It is used in the program for storing a value. 5. A program designed to handle 1000 simultaneous users, crashed when 1001 the user logged in. 6. A “while” loop never exits 7. User interface displays “MALFUNCTION 54” when something goes wrong in the back-end 8. No documentation (comments) for the source code 9. Hungarian Notation not followed while coding, even though the coding guide lines mandate to use Hungarian Notation 10. Pressing of “Tab” key moves the cursor in different fields of a web form randomly. <table><thead><tr><th>Statement</th><th>Defect Category</th><th>Defect Name</th></tr></thead><tbody><tr><td>1</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td></tr><tr><td>3</td><td></td><td></td></tr><tr><td>4</td><td></td><td></td></tr><tr><td>5</td><td></td><td></td></tr><tr><td>6</td><td></td><td></td></tr><tr><td>7</td><td></td><td></td></tr><tr><td>8</td><td></td><td></td></tr><tr><td>9</td><td></td><td></td></tr><tr><td>10</td><td></td><td></td></tr></tbody></table>	Statement	Defect Category	Defect Name	1			2			3			4			5			6			7			8			9			10			Unit – 7
Statement	Defect Category	Defect Name																																	
1																																			
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21.	Perform unit testing using JUNIT(Or any other testing tool).	Unit – 7																																	
22.	Consider the scenario of development of software for Travel Management System (TMS) is in progress. The TMS software has 3 major modules namely Ticket_Booking_Module, Hotel_Booking_Module and Taxi_Booking_Module.The Ticket_Booking_Module has 3 sub modules namely Enquiry_Module,Booking_Module andUpdate_Module. The enquiry module uses Date_Validation_Unit,Ticket_Validation_Unitand Place_Validation_Unit. <pre>graph TD TMS[Travel_Management_System] --> TBM[Ticket_Booking_Module] TMS --> HBM[Hotel_Booking_Module] TMS --> TBM[Taxi_Booking_Module] TBM --> EM[Enquiry_Module] TBM --> BM[Booking_Module] TBM --> UM[Update_Module] EM --> VM[View_Module] EM --> EdM[Edit_Module] EM --> CM[Cancel_Module] VM --> DVU[Data_Validation_Unit] EdM --> PVU[Place_Validation_Unit] CM --> TVU[Ticket_Validation_Unit] DVU --> SVU[Source_Validation_Unit] DVU --> DUU[Destination_Validation_Unit] PVU --> DUU TVU --> DUU</pre> In the context of the given scenario, identify the usage of stub or driver for the following situations. 1. Except the Ticket_validation_Unit, the coding and unit testing of all other modules, sub modules and units of TMS are completed. The top-down integration is in progress for the TMS software. To carry out the integration testing, which among the following is necessary? a) A Stub for Ticket_Validation_Unit b) A Driver For Ticket_Validation_Unit c) A Stub for Enquiry_Module d) A Driver for Enquiry_Module e) A Stub For Ticket_Booking_Module f) A Driver For Ticket_Booking_Module 2. The coding and unit testing of all the module, sub modules and units of TMS are completed except the Update_Module (coding and testing for Edit_Module, Cancel_Module and View_Module are also completed). The bottom-up integration is to be started for the TMS software. Mention any stub or driver needed to carry out the integration testing? 3. Except the Taxi_Booking_Module, the coding and unit testing of all other modules, sub modules and units of TMS are completed. The top-down integration is to be started for the TMS software. Mention any stub or driver needed to carry out the integration testing?	Unit – 7																																	
27.	Design different test cases for your chosen Project	Unit – 8																																	
28.	Draw CFG for following problems and show statement coverage, branch coverage and path coverage for each. Also Calculate cyclomatic complexity. Problem 1:- IF A = 354 THEN IF B > C THEN A = B ELSE A = C END IF END IF PRINT A Problem 2:- { int i, j, k; for (i=0 ; i<=N ; i++) p[i] = 1; for (i=2 ; i<=N ; i++) { k = p[i]; j=1;	Unit – 8																																	

	<pre> while (a[p[j-1]] > a[k] { p[j] = p[j-1]; j--; } p[j]=k; } Problem 3:- begin int x, y, power; float z; input(x, y); if(y<0) power = -y; else power = y; z=1; while(power!=0) { z=z*x; power=power-1; } if(y<0) z=1/z; output(z); end </pre>	
29.	Perform version control using GIT(Or any other version control tool).	Unit – 10

Course Outcome	
	<i>Upon completion of the course students will be able to</i>
CO1	Analyze and specify software requirements and apply various software process models to real-world software development scenarios, understanding their advantages and limitations.
CO2	Learn professional responsibilities associated with requirement analysis, function-oriented and object-oriented design techniques to develop structured and modular software solutions.
CO3	Develop comprehensive skills in managing software projects efficiently, including metrics application, estimation, scheduling, testing proficiency, and quality assurance understanding.
CO4	Develop comprehensive expertise in software testing, quality assurance, and change management, fostering a culture of continuous improvement.
Suggested Reference Books	
1	Software Engineering: A practitioner's approach (6 or 7 th Edition), Roger S. Pressman, McGraw Hill.
2	Fundamentals of Software Engineering (4 th Edition), Rajib Mall, Prentice Hall India.
3	Software Engineering, Ian Sommerville, Addison and Wesley

List of Open Source Software/Learning website	
1	https://www.javatpoint.com/
2	https://www.tutorialspoint.com/
3	https://www.guru99.com/
4	https://support.microsoft.com/en-us/office/beginner-tutorial-for-visio-bc1605de-d9f3-4c3a-970c-19876386047c
5	https://www.softwaretestingmaterial.com/manual-testing-tutorial/

Practical Project/Hands on Project																													
Sr. No.	Project List	Linked with Unit																											
1	Assume that you are Software Architect or Project Manager in organization. You have been assigned the task of constructing a website for a specific company with your team. Estimate project efforts, total development time and no of persons involved in your project using any one project estimation technique After completion of estimation as a project manager, would you quote the cost estimated using COCOMO as the price in your bid? Explain your answer.	Unit-6																											
2	The following table indicates the various tasks involved in completing a software project, the corresponding activities, and the estimated effort for each task in person-months. <table border="1"> <thead> <tr> <th>Notation</th><th>Activity</th><th>Efforts</th></tr> </thead> <tbody> <tr> <td>T1</td><td>Requirements specification</td><td>1</td></tr> <tr> <td>T2</td><td>Design</td><td>3</td></tr> <tr> <td>T3</td><td>Code actuator interface module</td><td>3</td></tr> <tr> <td>T4</td><td>Code sensor interface module</td><td>6</td></tr> <tr> <td>T5</td><td>Code user interface part</td><td>4</td></tr> <tr> <td>T6</td><td>Code control processing part</td><td>2</td></tr> <tr> <td>T7</td><td>Integrate and Test</td><td>7</td></tr> <tr> <td>T8</td><td>Write user manual</td><td>4</td></tr> </tbody> </table> The precedence relation $T_i \leq \{T_j, T_k\}$ implies that the task T_i must complete before either task T_j or T_k can start. The following precedence relation is known to hold among different tasks: $T_1 \leq T_2 \leq \{T_3, T_4, T_5, T_6\} \leq T_7$. (a) Draw the Activity network representation of the tasks. (b) Determine ES, EF and LS, LF for every task. (c) Develop the Gantt chart representations for the project.	Notation	Activity	Efforts	T1	Requirements specification	1	T2	Design	3	T3	Code actuator interface module	3	T4	Code sensor interface module	6	T5	Code user interface part	4	T6	Code control processing part	2	T7	Integrate and Test	7	T8	Write user manual	4	Unit-6
Notation	Activity	Efforts																											
T1	Requirements specification	1																											
T2	Design	3																											
T3	Code actuator interface module	3																											
T4	Code sensor interface module	6																											
T5	Code user interface part	4																											
T6	Code control processing part	2																											
T7	Integrate and Test	7																											
T8	Write user manual	4																											
3	Design the black-box test suite for the following Library Automation Software. The Library Automation Software accepts a string representing the enrollment no of a student. It checks the student’s account, and displays the details of book the student has issued. If the book is overdue then it displays the due date along with the fine to be paid.	Unit-8																											
4	Objective: To identify the various elicitation techniques and their usage for the Banking case study.	Unit-3																											

Background: Requirement elicitation is the process of seeking, discovering, acquiring and elaborating requirement. This includes learning and understanding the needs of the users. This activity is communication centric and iterative in the nature. The techniques used here are the important to get stack holder consensus on the requirements.

Problem Description:

KHL is a leading global bank that provides standard banking services to its customers spanning across the globe. The head office is located in London and the bank has presence in more than 20 countries with client base of nearly 500,000. Tuning with times and ever increasing clients and transactions, the bank has specialized branches for specific customer segments like consumer, corporate and the SME’s. KHL Bank aims to be one stop shop for its customers to address their changing financial needs. KHL bank offers various banking products and services across its customer segments including Core Banking and Wealth Management amongst other services. KHL Bank is well known among its clients for world-class processes and speed of execution of transactions as part of core banking. Currently, KHL bank has made a proposal for investing around \$200 million in setting-up 24x7 banking support facilities for the customers. The bank has decided to leverage IT for automating several of the business processes including:

- Managing Accounts
- Transaction Management

The aim of this proposed banking system is to create a paperless bank there by moving towards e-banking. FinSoft, a newly established software company has the vision of providing software solutions in the financial sector. Managing Director (MD) of KHL bank has approached FinSoft for the computerization of the bank so that there is no more manual way of doing transactions in any of its branches. As part of automation, the KHL bank users are to be provided with ATM facility, e-banking facility over internet and phone banking facility over land lines and cellular networks. FinSoft is doing such a project for the first Time. Requirements development team in FinSoft has planned for carrying out the requirement elicitation for this project.

In the context of the case study, for the following scenarios identify the most appropriate requirements elicitation techniques (Brainstorming, Workshops, Questionnaire, Task Analysis, Observation, Prototyping, Scenario identification).

Scenario	Requirement elicitation technique
Interrogative conversation with Managers, Cashiers, Clerks and other Staff for arriving at the requirement for automating transactions.	
Formal and planned requirement discussion in a conference to room conducted among managers of diversified branched facilitate by anchor.	
Survey form circulated among the users (account holders) who visit the bank, to ease their interactions with bank	
Analysis for understanding mode of transactions- Checks, Cash, DD, MT, Gold, etc.	
Ethnographers deployed for understanding the users interactions with bank officials.	
UI design of e-banking portal, ATM, Computer Systems	
Understanding the process involved in each transaction like withdraw, deposit, fund transfer etc.	

LOK JAGRUTI UNIVERSITY (LJU)

INSTITUTE OF ENGINEERING AND TECHNOLOGY

Department of CE/IT/CSD/AIML/AIDS /CSE/CST/CS&IT/CEA/MA&CP/CSE(AI)/ CSE(DS)/CSE(CS)

(701,702,703,704,705,712,713,714,715,716,717,718,719)

Bachelor of Engineering (B.E.) – Semester – I

Course Code:	117011192/117021192/117031192/117041192/117051192/117121192/117131192/117141192/117151192/017161192/017171192/017181192/017191192	Teaching Scheme				
Course Name:	Physics	Lecture (L)	Tutorial (T)	Practical (P)	Credit	Total Hours
Category of Course:	Basic Science Course (BSC)	3	0	2	4	30
Prerequisite Course:	---					

Syllabus				
Unit No.	Topic	Prerequisite Topic	Successive Topic	Teaching Hours
01	Dielectrics			3 (10%)
	1.1 Basic terms related to Dielectrics	---	---	
	1.2 Polarization mechanisms (No derivations)	Basic terms related to dielectrics (117011192 -Unit-1.1)	---	
	1.3 Internal / local field (No derivations) and Clausius-Mosotti equation	Polarization mechanisms (117011192 - Unit-1.2)	---	
	1.4 Types of dielectric materials	---	---	
	1.5 Applications (Capacitors & Transformers)	---	Charging and discharging of capacitor (117012291 – Unit-1.4)	
02	Magnetic Materials			3 (10%)
	2.1 Basic terms related to magnetism	---	---	
	2.2 Magnetic moment (No derivations)	Basic terms related to magnetism (117011192 -Unit-2.1)	---	
	2.3 Types of magnetic materials	---	---	
	2.4 Hysteresis and Hard and Soft Magnetic materials (Domain theory not included)	Basic terms related to magnetism (117011192 -Unit-2.1)	---	
	2.5 Applications (Magnetic hard disc)	---	---	
03	Ultrasonics			3 (10%)
	3.1 Definition and properties	---	---	
	3.2 Generation methods	---	---	
	3.3 Detection methods	Generation methods (117011192 -Unit-3.2)	---	
	3.4 Applications (Welding, drilling/cutting, SONAR, cleaning)	---	---	
04	Laser			3 (10%)
	4.1 Characteristics of Laser	---	---	
	4.2 Einstein's theory	---	---	
	4.3 Basic concepts related to Laser	Einstein's theory (117011192 -Unit-4.2)	---	
	4.4 Types of Laser	Basic concepts related to Laser (117011192 -Unit-4.3)	---	
	4.5 Applications (Laser in material processing, cutting, holography)	---	---	
05	Fibre Optics			3 (10%)
	5.1 Introduction	---	---	
	5.2 Advantages of Fibre optic system over conventional system	---	---	
	5.3 Construction and basic principle	---	---	
	5.4 Acceptance angle and Numerical aperture	Construction and basic principle (117011192 -Unit-5.3)	---	
	5.5 Types of fibre optic cables (V-number not to be included)	---	---	
	5.6 Fibre optic communication link	---	---	
06	Semiconductors			3 (10%)
	6.1 Properties and types of Semiconductors	---	---	
	6.2 Carrier statistics in Semiconductors (Only intrinsic semiconductors to be covered)	Properties and types of Semiconductors (117011192 -Unit-6.1)	---	
	6.3 Carrier Generation (creation of EHP) and carrier recombination	Properties and types of Semiconductors (117011192 -Unit-6.1)	---	
	6.4 Carrier Transport (drift and diffusion)	---	---	
07	Basic Diodes			3

	7.1 PN junction diode	Properties and types of Semiconductors (117011192 -Unit-7.1)	Basic idea about forward bias, reverse bias of Diode and VI characteristics (117012291 – Unit-8.1)	(10%)
	7.2 Energy band diagrams of PN junction diode	PN junction diode (117011192 -Unit-7.1)	---	
	7.3 Avalanche and Zener breakdown	---	---	
	7.4 Zener diode	---	---	
08	Digital Circuits			3 (10%)
	8.1 Basic gates AND, OR, NOT, NAND, NOR, EX-OR, EX-NOR, Universal Gates	---	---	
	8.2 Boolean algebra	---	---	
	8.3 Building AND, OR Gate with diodes	PN Junction Diode (117011192 -Unit-7.1)	---	
09	Measurements			2 (7%)
	9.1 Four probe method	---	---	
	9.2 Hall effect	---	---	
	9.3 Van der Pauw method	---	---	
	9.4 Hot probe method	---	---	
10	Special Purpose Diodes			4 (13%)
	10.1 Light emitting diode	Carrier Generation (creation of EHP) and carrier recombination (117011192 -Unit-6.3)	---	
	10.2 LDR	---	---	
	10.3 Photodiode	Carrier Generation (creation of EHP) and carrier recombination (117011192 -Unit-6.3)	---	
	10.4 Photovoltaic effect and PV cell (solar cell)	---	---	
	10.5 Metal-semiconductor diodes	---	---	

Sr No.	Practical Title	Link to Theory Syllabus
1	To find out the velocity & compressibility of liquid using Ultrasonic Interferometer.	Unit-3
2	To find out the wavelength of LASER source using Diffraction.	Unit-4
3	To find out the Numerical Aperture of a Fiber optic cable.	Unit-5
4	Understanding the working of PN Junction diode & plotting the V-I characteristics.	Unit-7
5	Study and plot the characteristics of Zener diode.	Unit-7
6	To find the bandgap of germanium PN Junction diode.	Unit-7
7	To verify truth table of AND, OR, NOT, NAND, NOR, EX-OR, EX-NOR gates	Unit-8
8	Understanding the characteristics of Photodiode and Solar cell.	Unit-9
9	To find the resistivity of semiconductor using Four probe method.	Unit-10
10	Study and plot the characteristics of Light emitting diode.	Unit-10

Major Components/ Equipment	
Sr. No.	Component/Equipment
1	Ultrasonic Interferometer, measuring cell, frequency generator, given liquid.
2	Optical bench, laser source, Optical screen, Double convex lens, Slit and Diffraction.
3	Patch Cords, Fibre optic Numerical Aperture measurement kit, 1 side connectorised ½ meter fibre cables of different diameters (1mm diameter plastic fibre & 0.5mm diameter plastic at 600nm).
4	PN Junction diode trainer kit.
5	Zener diode trainer kit
6	Diode trainer kit, mercury thermometer.
7	. Logic gates trainer kit
8	Four probe apparatus, sample (a Ge crystal in form of a chip), oven, thermometer, constant power supply, oven power supply, panel meters for measurement of current and voltage.
9	LED trainer kit.
10	Photodiode trainer kit, bulb, Solar cell trainer kit, solar panel, bulb.

Course Outcomes	
	<i>Upon completion of the course students will be able to</i>
CO 1	Demonstrate a thorough understanding of dielectrics, magnetic materials, and ultrasonics, including their properties, applications, and underlying principles in engineering and technology.
CO 2	Demonstrate proficiency in the principles and applications of laser and fiber optics technologies, enabling them to analyze advanced optical systems for various engineering endeavors.
CO 3	Have a thorough grasp of semiconductor physics, fundamental diode functionalities, and principles of digital circuit design, enabling them to proficiently analyze and resolve issues in electronic circuits
CO 4	Exhibit competence in measurement techniques and specialized diode applications, enhancing their capacity to employ accurate measurement methods and effectively utilize special purpose diodes
Suggested Reference Books	
1	Engineering Physics by G Vijayakumari, Vikas Publication
2	Engineering Physics by V Rajendran, Mc Graw Hill Education
3	Engineering Physics by Dattu Joshi, Mc Graw Hill Education
4	Engineering Physics by Dr. Rakesh Dogra, Katson books
5	Solid State Physics by S O Pillai, New Age International publishers

List of Open Source Software/Learning website	
1	http://nptel.ac.in/

Practical Project/Hands on Project		
Sr. No.	Practical List	Linked with Unit
1	Design a system showing uninterrupted power regulation in process industry.	Unit 01
2	Design an electric bell for an office.	Unit 02
3	Water enters a typical garden hose of diameter 1.6 cm with a velocity of 3 m/s. Calculate the exit velocity of water from the garden hose when a nozzle of diameter 0.5 cm is attached to the end of the hose. Also specify, what will be the divergence in this case.	Unit 03
4	Design a piezoelectricity-based visitor sensing doormat.	Unit 04
5	Design a basic laser security alarm system for locker system.	Unit 05
6	Design Advance data transfer system for beacon communication system in aircrafts.	Unit 06
7	Design electric protection circuit for power station.	Unit 07, 08, 10
8	Design an Automatic street light.	Unit 08, 09, 10

LOK JAGRUTI UNIVERSITY (LJU)

INSTITUTE OF ENGINEERING & TECHNOLOGY

Department of CE/IT/CSD/AIML/AIDS /CSE/CST/CS&IT/CEA/MA&CP/CSE(AI)/ CSE(DS)/CSE(CS)

(701,702,703,704,705, 712,713,714,715,716,717,718,719)

Bachelor of Engineering (B.E.) – Semester – I

Course Code:	017012192/017022192/017032192/017042192/017052192/017122192/017132192/017142192/017152192/017162192/017172192/017182192/017192192	Teaching Scheme			
Course Name:	IOT Workshop - Laboratory	Lecture (L)	Tutorial (T)	Practical (P)	Credit
Category of Course:	Engineering Science Course (ESC)	0	0	4	2
Prerequisite Course:	---				

Sr No.	Practical Title	Link to Theory Syllabus
1	Understating PROTEUS, ARDUINO (Nano/ Uno/ Mega), NODEMCU/ESP32	--
2	IR & Ultrasonic Sensor Interfacing with ARDUINO /NODEMCU/ESP32 & Simulation with PROTEUS	--
3	PIR Sensor Interfacing with ARDUINO /NODEMCU/ESP32 & Simulation with PROTEUS	--
4	Gas Sensor & Flame sensor Interfacing with ARDUINO /NODEMCU/ESP32 & Simulation with PROTEUS	--
5	LM35 Interface Interfacing with ARDUINO /NODEMCU/ESP32 & Simulation with PROTEUS	--
6	Moisture Sensor & DHT11/22 Interfacing with ARDUINO /NODEMCU/ESP32 & Simulation with PROTEUS	--
7	2-Channel Relay Interfacing with ARDUINO /NODEMCU/ESP32 & Simulation with PROTEUS	--
8	Display (16x 2/ 16x x4 LCD) Interfacing with ARDUINO /NODEMCU/ESP32 & Simulation with PROTEUS	--
9	Working on soldering with GPP	--
10	MINI PROJECT	--

Major Components/ Equipment		
Sr. No.	Component/Equipment	Specification
1	PROTEUS Software	
2	Controllor board: ARDUINO (Nano/ Uno/ Mega), NODEMCU/ESP32	Microcontroller ATmega328. Operating Voltage (logic level): 5 V. Input Voltage (recommended): 7-12 V. Input Voltage (limits): 6-20 V. Digital I/O Pins : 14 (of which 6 provide PWM output) Analog Input Pins: 8. DC Current per I/O Pin: 40 mA.
3	IR Sensor	VCC: External 3.3V-5V voltage (can be directly connected to 5v MCU and 3.3v MCU) GND: GND External OUT: Small board digital output interfaces (0 and 1)
4	Ultrasonic Sensor	Transmitter & Receiver Technology Used Non-Contact Technology Operating Voltage 5 V Operating Frequency 4 MHz Detection Range 2cm to 400cm Measuring Angle 30° Resolution 3mm Operating Current <15mA Sensor Dimensions 45mm x 20mm x 15mm
5	PIR Sensor	The HC-SR501 Operating Voltage 5 V to 20 V Current consumption: 65 mA Output Voltage: 3.3 V on condition Delay time: 5 seconds to 5 minute. Sensitivity Range: 3 meter to 7 meters
6	Gas Sensor	MQ2 Operating voltage: 5V Load resistance: 20 KΩ Heater resistance: 33Ω ± 5% Heating consumption: <800mw Sensing Resistance: 10 KΩ – 60 KΩ Concentration Scope: 200 – 10000ppm Preheat Time: Over 24 hour
7	Flame sensor	Operating Voltage: 3.3V to 5V DC Operating Current: 15ma

		Output Digital - 0V to 5V, Adjustable trigger level from preset Output Analog - 0V to 5V based on infrared radiation from fire flame falling on the sensor LEDs indicating output and power PCB Size: 3.2cm x 1.4cm LM393 based design
8	LM35	Calibrated directly in Celsius (Centigrade) Linear + 10.0 mV/ C scale factor 0.5 C accuracy guaranteeable (at +25 C) Rated for full -55 to +150 C range Suitable for remote applications Low cost due to wafer-level trimming Operates from 4 to 30 volts Less than 60 A current drain Low self-heating, 0.08 C in still air Nonlinearity only 1/4 C typical Low impedance output, 0.1 W for 1 mA load
9	Moisture Sensor	Operating Voltage 3.3V-5V. Module Dual Output mode, a simple digital output, and analog output more accurate. With fixed bolt hole for easy installation. Small PCB board size: 3cm * 1.6cm. Power indicator (red) and the digital switch output indicator (green). Using LM393 comparator chip, stable. VCC external 3.3V-5V GND GND External DO small board digital output interfaces (0 and 1) AO small board analog output interface
10	DHT11/22 Sensor	Super compact size Super low power consumption Super low voltage operation Standard I2C and 1-wire interface. Semi-conductor technology Sensing range Temperature: -20 ~ +60 C Humidity: 20-95 RH Humidity: Resolution: 0.1%RH Repeat: +- 1%RH Precision 25C @ +-5RH Temperature: Resolution: 0.1C Repeat: +-0.2C Precision: 25C @ +-0.5C Power: DC 2.7-5.5V Normal current 1mA Standby current 60uA Sample cycle: > 2 seconds Pin interface: 1. VDD 2. SDA 3. GND 4. SCL (connect to GND when use as 1-wire)
11	Relay	2-Channel, 5 A, 230V
12	LCD	16x 2/ 16x4
13	Soldering iron along with soldering flux & wax, De-soldering pump, standard size GPP	-

Course Outcome	
	<i>Upon completion of the course students will be able to</i>
CO1	Learning usage of tools for IOT environment
CO2	Learn sensor Interfacing with various controller boards for IOT application.
CO3	Understanding applications of various Sensors
CO4	Learning Hardware Programming with Microcontroller.
Suggested Reference Books	
1	Beginning Arduino, Michael McRobetr's Technology in Action
2	Exploring Arduino, Jeremy Blum. Wiley
3	NodeMCU ESP8266 Communication Methods and Protocols : Programming with Arduino IDE, Manoj R. Thakur

List of Open Source Software/Learning website	
1	http://arduino.cc
2	www.instructables.com/id/Arduino-Projects/
3	http://www.jeremyblum.com/category/arduino-
4	https://www.labcenter.com/downloads/
5	https://rntlab.com/learn-esp32-welcome/

Practical Project/Hands on Project		
Sr. No.	Project List	Linked with Unit
1	Design Mini Weather Station using Arduino Uno/NODEMCU.	--
2	Real time Data Logger Using Arduino Uno/NODEMCU.	--
3	Smart Home Automation Using Arduino Uno/NODEMCU.	--
4	Smart Irrigation System Using Arduino Uno/NODEMCU.	--
5	Health Monitoring System Using Arduino Uno/NODEMCU.	--
6	Advance Fire Alarm System Using Arduino Uno/NODEMCU.	--
7	Smart Room Temperature Using Arduino Uno/NODEMCU.	--

LOK JAGRUTI UNIVERSITY (LJU)
INSTITUTE OF ENGINEERING & TECHNOLOGY

Department of CE/IT/CSD/AIML/AIDS /CSE/CST/CS&IT/CEA/MA&CP/CSE(AI)/ CSE(DS)/CSE(CS)

(701,702,703,704,705 ,712,713,714,715,716,717,718,719)

Bachelor of Engineering (B.E.) – Semester – I

Course Code:	017012193/017022193/017032193/017042193/017052193/017122193/017132193/017142193/017152193/017162193/017172193/017182193/017192193
Course Name:	Computer Workshop - Laboratory
Category of Course:	Engineering Science (ESC)
Prerequisite Course:	---

Teaching Scheme			
Lecture (L)	Tutorial (T)	Practical (P)	Credit
0	0	4	2

Sr No.	Practical Title	Link to Theory Syllabus
1	Introduction to various components of computer, Input / Output devices.	--
2	BIOS/UEFI settings	--
3	Basics and Installation of operating system, concept of dual booting, virtualization	--
4	Installation of device drivers and other required software, need and method of backup	--
5	Basic security issues: computer viruses, malware, etc., anti-virus software	--
6	Various office tools for text, spreadsheet and presentation documents	--
7	Introduction to various online tools for forms, documents, storage, etc.	--
8	Basic concepts of networking and resource sharing in a network	--

Major Components/ Equipment		
Sr. No.	Component/Equipment	Specification
1	Computer	
2	Software's	MS Office, IP Messenger, Windows OS

Course Outcome	
	<i>Upon completion of the course students will be able to</i>
CO1	Understand basics of computer hardware
CO2	Install operating systems, drivers related to hardware and other software
CO3	Create reports, presentation, and spread sheets
CO4	Manage security related basic problem using anti-virus software.
Suggested Reference Books	
1	Scott Mueller, “Upgrading and repairing PCs”, 21st Edition; Pearson Education.
2	Stephen J Bigelow, “Troubleshooting, Maintaining and Repairing PCs”, 5th Edition; Tata McGrawHill Publication
3	M.Radhakrishnan and D. Balasubramanian, “Computer Installation and Troubleshooting” ISTE Learning Material.

List of Open Source Software/Learning website	
1	www.youtube.com
2	www.wikihow.com
3	www.tomsguide.com
4	www.tutorialspoint.com
5	www.makeuseof.com

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(701,702,703,704,705 ,712,713,714,715,716,717,718,719)

Bachelor of Engineering (B.E.) – Semester – I

Course Code:	017018191/017028191/017038191/017048191/017058191/017128191/ 017138191/017148191/017158191/017168191/017178191/017188191/ 017198191
Course Name:	Environmental Science
Category of Course:	Mandatory Course (MC)
Prerequisite Course:	---

Teaching Scheme				
Lecture (L)	Tutorial (T)	Practical (P)	Credit	Total Hours
2	0	0	0	20

Syllabus				
Unit No.	Topic	Prerequisite Topic	Successive Topic	Teaching Hours
01	Introduction to Environment			1 (5%)
	1.1 Definition, principles and scope of Environmental Science	---	---	
	1.2 Impacts of technology on Environment, Environmental Degradation,	---	---	
	1.3 Importance for different engineering disciplines	---	---	
02	Water Pollution			2 (10%)
	2.1 Introduction – Water Quality Standards	---	---	
	2.2 Sources of Water Pollution	---	---	
	2.3 Classification of water pollutants	---	---	
	2.4 Effects of water pollutants	---	---	
03	Air Pollution			2 (10%)
	3.1 Composition of air	---	---	
	3.2 Structure of atmosphere	---	---	
	3.3 Ambient Air Quality Standards	---	---	
	3.4 Classification of air pollutants	---	---	
	3.5 Sources of common air pollutants like PM, SO ₂ , NO _X , Auto exhaust	---	---	
	3.6 Effects of common air pollutants	---	--	
04	Noise Pollution			2 (10%)
	4.1 Introduction	---	---	
	4.2 Sound and Noise	---	---	
	4.3 Noise measurements	---	---	
	4.4 Causes and Effects	---	---	
05	Solid waste management			2 (10%)
	5.1 Introduction	---	---	
	5.2 Types and Sources	---	---	
	5.3 Cause and Effect	---	---	
	5.4 Solid waste Management: Collection ,Processing ,Disposal	---	---	
06	Biomedical waste management			2 (10%)
	6.1 Introduction	---	---	
	6.2 Sources	---	---	
	6.3 Classification	---	---	
	6.4 Management: Segregation, Transportation, Treatment	---	---	
07	Electronic Waste Management			2 (10%)
	7.1 Introduction	---	---	
	7.2 Classification, Generation of Waste	---	---	
	7.3 International Trade or E-waste Dumping in Developing countries	---	---	
	7.4 Impacts of E-waste on Environment and Human Health	---	---	
	7.5 Management of E-waste	---	---	
08	Global Environmental Issue			3 (15%)
	8.1 Introduction	---	---	
	8.2 Climate Change	---	---	
	8.3 Greenhouse and Global Warming	---	---	
	8.4 Acid rain	---	---	
	8.5 Ozone Depletion	---	---	
	8.6 Carbon Foot Print	---	---	

	8.7 Benefits of Carbon foot prints	---	---	
	8.8 Cleaner Development Mechanism	---	---	
	8.9 International Steps for mitigation Global change	---	---	
09	Green Technologies			2 (10 %)
	9.1 Design	---	---	
	9.2 Operational Parameters	---	---	
	9.3 Maintenance	---	---	
	9.4 Solar Energy	---	---	
	9.5 Wind Energy	---	---	
	9.6 Biomass Energy	---	---	
10	Social issues and Environment			2 (10%)
	10.1 Unsustainable to Sustainable Development	---	---	
	10.2 Urban problems related to energy	---	---	
	10.3 Population Growth, Impact of Population, Gender and Environment	---	---	
	10.4 Role of individual to protect Environment	---	---	
	10.5 Role of information Technology to protect Environment and Human health	---	---	

Course Outcome	
	<i>Upon completion of the course students will be able to</i>
CO1	Develop the ability to identify various types of pollution such as air pollution, water pollution prevalent in society, comprehensively understanding their sources and the consequential impacts on both human health and the environment.
CO2	Develop the ability to identify noise pollution, their sources and effect on human and environment. Acquire an in-depth understanding of different solid and bio medical waste management strategies and their crucial significance in preserving both human health and the environment.
CO3	Understand various futuristic challenges such as e-waste and critical issue related to climate change, gaining insights into global initiatives and efforts aimed at addressing this critical environmental challenge.
CO4	Examine the role of eco-friendly technology in fostering sustainable development, considering both environmental and social implications.
Suggested Reference Books	
1	Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha Second edition, 2013 Publisher: Universities Press (India) Private Ltd, Hyderabad
2	Basics of Environmental Studies by U K Khare, 2011 Published by Tata McGraw Hill
3	Environmental Science by B.R Shah and Dr.Sneha Popli Mahajan Publication House
4	Environmental Sciences by Daniel B Botkin & Edward A Keller Publisher: John Wiley & Sons.
5	De A.K., Environmental Chemistry, Wiley Eastern Ltd.
6	Agarwal, K.C.2001 Environmental Biology, Nidi Publ.Ltd.Bikane.
7	Renewable Energy and Technology by DR.P.Subrahmanian and DR.A.Sampatharajan

List of Open Source Software/Learning website	
1	https://www.coursera.org/browse/physical-science-and-engineering/environmental-science-and-sustainability
2	https://www.classcentral.com/course/swayam-environmental-pollution-and-global-issues-22968
3	https://www.edx.org/learn/renewable-energy
4	https://www.coursera.org/learn/solid-waste-management
5	https://www.udemy.com/course/basic-medicalbiomedical-waste-management-course/
6	https://onlinecourses.nptel.ac.in/noc20_ce12/preview